

Supply Chain & Inventory Analytics

Full Project Report

Data Analyst Portfolio Project

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Tools: Python, Pandas, MySQL, Excel, Power BI, DAX

End-to-end analysis of inventory, orders, products, suppliers, procurement and warehouse operations.

1. Executive Summary

This project presents an end-to-end Data Analyst workflow for a supply chain and inventory business environment. Operational data covering products, suppliers, inventory, customer orders and procurement was cleaned using Python and Pandas, analyzed using MySQL, prepared in Excel, and visualized through a three-page Power BI dashboard.

The analysis focuses on inventory efficiency, stockout risk, product demand, warehouse performance, supplier fulfillment, procurement spending, supplier lead time, late deliveries and operational fulfillment. The final solution converts raw operational data into structured KPIs and business recommendations.

2. Business Objectives

- Monitor inventory levels and identify stockout risks.
- Analyze product demand and sales performance.
- Compare warehouse performance.
- Evaluate supplier fulfillment and delivery performance.
- Analyze procurement quantities and procurement costs.
- Measure supplier lead time and late-delivery rates.
- Create interactive management KPIs and dashboards.
- Generate actionable recommendations from operational data.

3. Technology Stack

Technology	Purpose
Python	Data processing and automation
Pandas	Data cleaning and feature engineering
MySQL	Business analysis and SQL queries
Excel	Data preparation and validation
Power BI	Interactive dashboard and visualization
DAX	Dynamic KPI calculations
GitHub	Portfolio presentation and version control

4. Dataset Overview

Five operational datasets were used in the project. The cleaned data was used consistently across SQL, Excel and Power BI analysis.

Dataset	Raw Rows	Cleaned Rows	Purpose
Products	100	100	Product information
Suppliers	25	25	Supplier information
Inventory	1,600	1,600	Stock and warehouse activity
Orders	10,000	9,489	Customer order activity
Procurement	2,000	1,988	Purchase and supplier activity

The reduction in Orders and Procurement rows reflects duplicate records removed during the cleaning process.

5. Data Cleaning & Transformation

Python and Pandas were used to create a reliable analytical dataset before business analysis.

- Loaded and inspected raw CSV files.
- Checked row counts, columns and data types.
- Identified and removed duplicate records.
- Handled data-type and date-related issues.
- Prepared analytical fields and derived metrics.
- Exported cleaned datasets for SQL, Excel and Power BI.
- Used a repeatable Python script for the data preparation workflow.

6. MySQL Business Analysis

The cleaned datasets were loaded into a MySQL database named **supply_chain**. The main tables are products, suppliers, inventory, orders and procurement.

SQL analysis covered:

- Inventory levels and stockout activity.
- Product and sales performance.
- Warehouse performance.
- Supplier performance.
- Procurement quantities and spending.
- Order and procurement fulfillment.
- Supplier lead time.
- Late-delivery analysis.
- Aggregations, joins, CASE expressions and date analysis.

7. Excel Preparation

The cleaned datasets were consolidated into an Excel workbook for validation, organization and reporting. The workbook contains Products, Suppliers, Inventory, Orders and Procurement data.

8. Power BI Dashboard

The final Power BI report contains three pages.

Page 1 — Supply Chain Executive Overview

- Total Sales
- Total Orders
- Total Units Sold
- Total Procurement Cost
- Stockout Rate
- Fulfillment Rate
- Monthly Sales Trend
- Sales by Category

- Sales by Warehouse
- Year, Warehouse and Category slicers

Page 2 — Inventory & Product Performance

- Total Closing Stock
- Total Damaged Units
- Inventory Turnover
- Stockout Rate by Warehouse
- Top 10 Products by Units Sold
- Closing Stock by Category
- Units Sold by Category
- Warehouse and Category slicers

Page 3 — Supplier & Procurement Performance

- Total Purchase Orders
- Total Quantity Ordered
- Total Quantity Received
- Procurement Fulfillment Rate
- Supplier Fulfillment Rate
- Top Suppliers by Procurement Spend
- Average Supplier Lead Time
- Late Delivery Rate by Supplier
- Procurement Cost by Warehouse
- Monthly Procurement Cost Trend
- Warehouse slicer

9. Key DAX Measures

Measure	Logic
Total Procurement Cost	SUM(Procurement[procurement_cost])
Total Purchase Orders	DISTINCTCOUNT(Procurement[procurement_id])
Total Quantity Ordered	SUM(Procurement[quantity_ordered])
Total Quantity Received	SUM(Procurement[quantity_received])
Procurement Fulfillment Rate	Quantity received ÷ quantity ordered
Average Lead Time	Average DATEDIFF between PO date and actual delivery
Late Deliveries	Count of records where actual delivery > expected delivery
Late Delivery Rate	Late deliveries ÷ total purchase orders

10. KPI Definitions

KPI	Definition
Total Sales	Total sales generated from customer orders.

KPI	Definition
Total Orders	Number of customer orders.
Total Units Sold	Total quantity sold.
Closing Stock	Inventory remaining at the end of the inventory period.
Stockout Rate	Percentage of inventory records with a stockout.
Inventory Turnover	Indicator of inventory movement efficiency.
Fulfillment Rate	Percentage of orders successfully fulfilled.
Procurement Fulfillment Rate	Quantity received divided by quantity ordered.
Average Lead Time	Average time between purchase order date and actual delivery.
Late Delivery Rate	Percentage of procurement orders delivered after expected date.
Procurement Cost	Total procurement expenditure.

11. Business Insights

Inventory and Stockout Risk — The dashboard helps identify warehouses and inventory areas experiencing higher stockout activity, supporting replenishment and safety-stock decisions.

Product Demand — Product-level analysis helps identify high-demand products that require closer inventory monitoring.

Supplier Performance — Supplier fulfillment metrics identify suppliers that consistently meet or fall short of ordered quantities.

Supplier Lead Time — Average lead time highlights suppliers requiring longer delivery periods and supports procurement planning.

Late Deliveries — Late-delivery analysis highlights suppliers and procurement activities where actual delivery occurs after the expected date.

Procurement Spending — Procurement cost analysis helps identify warehouses and suppliers with higher procurement expenditure.

Warehouse Performance — Warehouse-level analysis provides a basis for comparing inventory, sales, stockouts and procurement activity.

Fulfillment Performance — Order and procurement fulfillment metrics provide visibility into operational execution.

12. Business Recommendations

- Monitor warehouses with higher stockout rates.
- Improve replenishment planning for high-demand products.
- Maintain appropriate safety-stock levels.
- Evaluate suppliers using fulfillment performance.
- Investigate suppliers with consistently high lead times.
- Investigate suppliers with high late-delivery rates.
- Monitor procurement spending by warehouse and supplier.
- Use historical demand patterns to improve procurement planning.
- Prioritize high-demand products to reduce stockout risk.
- Use supplier performance metrics during procurement decisions.

13. Project Structure

```
Supply_Chain_Inventory_Analytics/  
■■■ README.md  
■■■ raw_data/  
■■■ cleaned_data/  
■■■ python/project.py  
■■■ sql/supply_chain_analysis.sql  
■■■ output/Supply_Chain_Analytics.xlsx  
■■■ screenshots/  
    ■■■ executive_overview.png  
    ■■■ inventory_product.png  
    ■■■ supplier_procurement.png
```

14. Skills Demonstrated

Python/Pandas: Data cleaning, transformation, duplicate handling, feature engineering and data export.

SQL/MySQL: Joins, aggregation, filtering, grouping, CASE expressions, date analysis and KPI calculations.

Excel: Data preparation, validation and reporting.

Power BI/DAX: Data modeling, relationships, measures, KPI cards, charts, slicers and interactive dashboards.

15. Future Improvements

- Demand forecasting
- Inventory demand prediction
- Supplier risk scoring
- ABC inventory analysis
- Safety-stock optimization
- Reorder-point optimization
- Machine Learning for demand forecasting
- Automated data pipelines

16. Conclusion

This project demonstrates a complete Data Analyst workflow from raw operational data to business decision-making. It combines Python/Pandas for data preparation, MySQL for business analysis, Excel for structured reporting, and Power BI/DAX for interactive visualization.

The resulting solution provides a practical view of inventory, products, orders, suppliers, procurement, warehouses and operational KPIs. The analysis can support better inventory planning, supplier evaluation, procurement decisions and warehouse management.

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